MATH317 · NUMERICAL METHODS

Midterm 462 Practice

18/03/2025 · 60 minutes · 30 marks

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01 Questions

Answer each question in the space below it. The full solution is in a separate file.

Q1 4 + 8 M

1. Given $x = 30.158$ rounded to $\tilde{x} = 30.16$, then:

a) Calculate the absolute error between the true value and the computed value.

b) Calculate the relative error as a percentage.

2. Given the polynomial $P(x) = 4x^3 - 8x^2 + 14x - 32$.

a) Use Horner's algorithm to evaluate $P(3)$.

b) Using the complete Horner's algorithm, find the Taylor expansion of $P(x)$ about the point $r = 2$.

Q2 7 M

Determine the double-precision machine representation in a 64-bit word-length computer of the decimal number (−184.125).

Q3 3 + 8 M

Given $f(x) = 2x^2 - e^x$.

1. In a bisection method, if you are asked to find a root in the interval [1, 2] with an accuracy level of $\varepsilon = 10^{-2}$, how many iterations does it take to find the root?
2. Find a root of f(x) on the interval [0, 2] by using the bisection method. Terminate the iterations if they exceed nmax = 2 or if the error bound is less than 0.5×10^{-6}.